

Hospital 30-Day Readmission Prediction

Predicting high-risk patients at discharge to target limited-capacity care management resources

CLINICAL ML · PORTFOLIO PROJECT

0.67

Best ROC-AUC (Logistic Regression) — in line with published readmission models (0.65–0.70)

38.7%

Precision in the top 10% highest-risk patients, vs. 19.2% base rate — 2x lift

\$918K

Est. annual net savings targeting just the top 5% risk tier at a 20K-discharge hospital

1.84x

ROI at the 5% capacity tier — the best return of any intervention capacity tested

MODEL COMPARISON (HELD-OUT TEST SET)

MODEL	ROC-AUC	PR-AUC	PREC @ TOP 10%
Logistic Regression BEST	0.670	0.322	38.7%
Random Forest	0.650	0.311	40.7%
XGBoost	0.641	0.290	34.7%

Note: the interpretable baseline edged out tree ensembles here — with EHR tabular features this size, that's common, and it's a win for clinical trust, not just a tie.

TOP RISK DRIVERS (SHAP)



SHAP (TreeExplainer, XGBoost) gives per-patient explanations too — care teams see why a patient was flagged, not just a score.

COST / IMPACT SIMULATION

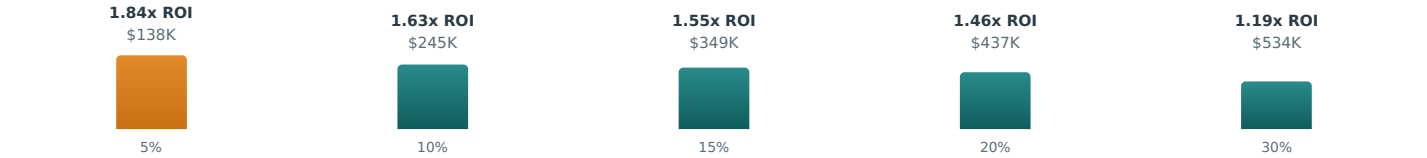
Targeted intervention beats broad coverage

Enrolling the top 5% highest-risk discharges in a transitional-care program (nurse follow-up, med reconciliation) — rather than casting a wider net — produced the best return, even though wider tiers prevented more total readmissions.

1.84x ROI

vs. 1.19x ROI at the 30% capacity tier

ROI VS. NET SAVINGS BY INTERVENTION CAPACITY



Bar height = ROI; labels show ROI and net savings at each capacity tier (% of discharges enrolled). ROI peaks narrow at 5%, but total dollars saved keep climbing with reach — a real staffing tradeoff, not a modeling one.

Assumptions: \$15,200 avg. cost per avoidable readmission (AHRQ/HCUPI) · \$500 per patient enrolled in transitional care · 25% relative risk reduction for enrolled patients (Coleman Care Transitions Intervention / Project RED literature) · Annualized estimate scales the 3,000-patient test cohort to a 20,000-discharge/year hospital.

KEY MODELING DECISIONS

Why PR-AUC over ROC-AUC

With ~19% positive class rate, ROC-AUC can look strong while precision at any realistic staffing threshold stays low. PR-AUC and precision@K reflect what matters operationally: how many true high-risk patients you catch within a fixed intervention capacity.

Why class weighting over resampling

Imbalance was handled via `class_weight="balanced"` rather than SMOTE-style oversampling, to keep predicted probabilities calibrated and trustworthy for risk-stratification rather than distorted by synthetic samples.

Why narrow targeting wins on ROI

Net savings kept rising as capacity widened from 5%→30%, but ROI peaked at the narrowest, highest-confidence tier — a real constraint given most care-management teams can't enroll every discharged patient.

HONEST LIMITATIONS & NEXT STEPS

Performance ceiling is expected. Published readmission models (including CMS's own) typically top out around 0.65–0.70 ROC-AUC — a large share of risk lives outside the EHR (housing, caregiver support, adherence), not in a modeling gap.

Before production deployment: validate on a temporal (not random) holdout, run a fairness audit across race/age/gender subgroups, and set the operating threshold jointly with the care-management team.

Stack: Python · pandas · scikit-learn · XGBoost · SHAP | Data: Synthetic EHR (schema mirrors UCI Diabetes 130-Hospitals) | Eval: ROC-AUC, PR-AUC, calibration, precision@K | Full pipeline, code, and evaluation plots available on request — reproducible end-to-end via a single script.